H–B–621f

December 29, 1958

SUPERSEDING
Int. Fed. Spec. H–B–OO621e (GSA–FSS)
February 24, 1958 and
Fed. Spec. H–B–621d
June 30, 1955

FEDERAL SPECIFICATION

BRUSH, STENCIL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers stencil-ink spreading bristle brushes primarily intended to be used to spread ink through stencils.

1.2 Classification.

1.2.1 *Types and sizes.*—Stencil brushes shall be of the following types and sizes, as specified (see 6.1):

Type L.—Long handle.
- Size ½-inch
- Size 13/16-inch
- Size 1 1/16-inch

Type S.—Short handle.
- Size ⅞-inch
- Size 1⅛-inch
- Size 1⅜-inch

Type F.—Fountain handle.
- Size 1⅜-inch

NOTE.—Size designation is same as dimensional requirement of diameter inside at bristle (see table I).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

R–N–91—Naphthalene; Balls and Flakes.
TT–B–838—Butyl Acetate; Normal (for Use in Organic Coatings).
TT–B–846—Butyl Alcohol; Normal (for Use in Organic Coatings).
TT–M–261—Methyl-Ethyl-Ketone (for Use in Organic Coatings).
TT–T–548—Toluol (for Use in Organic Coatings).
PPP–B–566—Boxes, Folding, Paperboard.
PPP–B–585—Boxes, Wood, Wirebound.
PPP–B–591—Boxes, Fiberboard, Wood-Cleated.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Boxes Fiber.
PPP–B–665—Boxes, Paperboard, Metal Stayed (including stay material).
PPP–B–676—Boxes, Set-Up, Paperboard.
PPP–T–60—Tape; Pressure Sensitive Adhesive, Waterproof, for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

102532—59

FSC 7520

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-B-4229—Boxes, Paperboard, Metal-Stayed.

MIL-B-10377—Boxes, Wood-Cleated, Veneer, Paper Overlaid.

MIL-L-10547—Liners, Case, Waterproof.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33d Street, New York 16, N. Y.

3. REQUIREMENTS

3.1 Standard samples (types L and S).—Types L and S brushes shall be equal to the standard samples in workmanship and balance; in stiffness, solidity and quality of bristle; and in any other respects not specifically covered in this specification. (See 6.1 and 6.2.)

3.2 Material.

3.2.1 *Hog bristle.*—Bristle shall be 100 percent hog bristle, boiled or steamed, straightened, and thoroughly mixed and cleaned. No riflings shall be used. The brushes shall be trimmed solid to a square top.

3.2.2 *Metals.*—Metals used in the fabrication of the stencil brushes shall be resistant to the corrosive effects of inks and cleaning solutions normally used for stenciling.

3.2.3 *Handles.*—The handles for types L and S brushes shall be close-grained hardwood, smoothly finished with varnish, lacquer or other suitable transparent organic finish.

3.2.3.1 *Coating.*—The coating (finishing) material for wood handles shall be of varnish, lacquer, or any other suitable transparent organic finish.

3.3 Set.—The butt end of all bristle shall be set either by vulcanizing in hard rubber or by setting in a suitable compound.

3.3.1 *Setting compound.*—When tested as specified in 4.5.2.2, the setting compound shall show no excessive softening; the bristles shall remain firmly embedded in the setting compound; and there shall be no evidence of solubility of the setting compound in the solvent. After the brush has been removed from the solvent and air-dried for 24 hours, the setting compound shall show no deterioration and there shall be no loosening of the bristles.

TABLE I.—*Minimum physical requirements*

	Type L			Type S			Type F
	Size ½-inch	Size 13/16-inch	Size 1 1/16-inch	Size ⅞-inch	Size 1⅛-inch	Size 1⅜-inch	Size 1⅝-inch
Length, over all---inches	9 1/16	9 1/16	9 1/16	4¼	4⅝	4⅞	7
Ferrule:							
Diameter, inside at bristle---inches	½	13/16	1 1/16	⅞	1⅛	1⅜	1⅝
Depth---inches	1¼	1¼	1½	1	1	1	----
Thickness---inch	0. 017	0. 020	0. 028	0. 035	0. 045	0. 045	----
Bristle:							
Length, inside of ferrule---inch	⅝	⅝	⅝	½	½	½	----
Length, outside of ferrule---inches	13/16	15/16	1 1/16	1⅛	1⅜	1½	1½
Weight---ounces	13/64	3/8	11/16	13/32	23/32	1 1/16	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

3.4 Identification marking.

3.4.1 *Nonmilitary.*—Each brush shall bear the manufacturer's name or trade-mark and the stock number stamped or otherwise permanently marked upon the item of equipment.

3.4.2 *Military.*—Brushes shall be identified in accordance with Military Standard MIL–STD–130.

3.5 Type L, long handle.

3.5.1 *Ferrule.*—The ferrule shall be nickel plated seamless steel and shall be pinned to the handle by at least three nails. The bottom edge of the ferrule shall have a crimped or rolled edge to retain the bristle setting.

3.5.2 *Physical requirements.*—Brushes shall conform to the applicable requirements of table I and shall be similar to figure 1.

3.6 Type S, short handle.

3.6.1 *Ferrule.*—The ferrule shall be as specified in 3.5.1.

3.6.2 *Set.*—The set shall be as specified in 3.3 and 3.3.1.

3.6.3 *Physical requirements.*—Brushes shall conform to the applicable requirements of table I and shall be similar to figure 2.

3.7 Type F, fountain handle.

3.7.1 *Design.*—Type F stencil brushes shall be comprised of a handle containing an ink reservoir, a brush tip, and a valve for manual control of ink flow from reservoir to brush tip.

3.7.2 *Handle.*—The handle shall be a one-piece metal shell (see 3.2.2). The reservoir formed by the shell shall hold not less than two fluid ounces of ink. Means shall be provided to facilitate refilling reservoir.

3.7.3 *Valve housing.*—The housing shall be threaded at one end to receive the brush tip.

3.7.4 *Valve.*—The valve shall be manually operated to open and return to closed position by spring action. The valve shall not leak in the closed position and valve actuating shaft shall be sealed to prevent leakage. Valve assembly shall be removable for cleaning or replacement. The valve outlet shall be within the valve housing and shall feed ink directly to the brush tip.

3.7.5 *Brush tip.*—The brush tip shall contain the bristles and shall be threaded to fit into the valve housing (see 6.4).

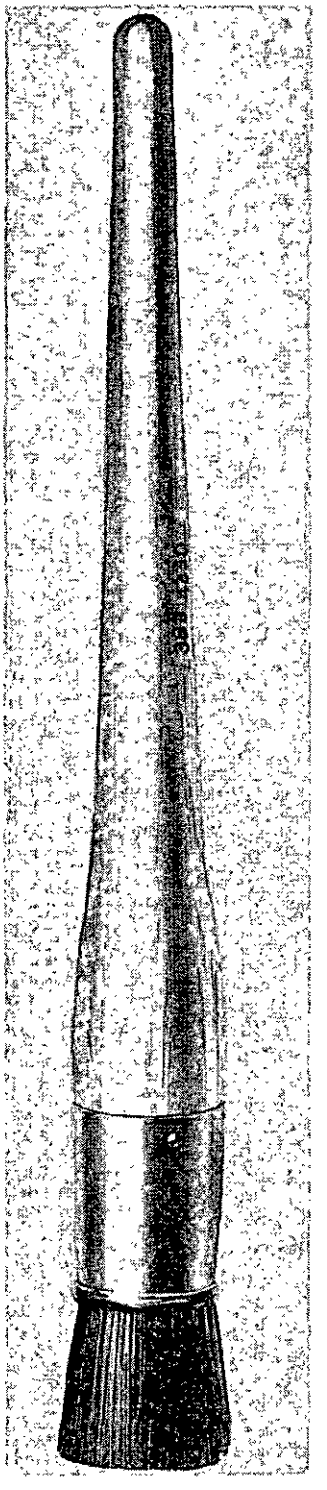


FIGURE 1.—TYPE L, LONG HANDLE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.



FIGURE 2.—TYPE S, SHORT HANDLE.

3.7.6 *Brush-tip cover.*—When specified (see 6.1), a brush-tip cover of light-weight metal shall be furnished. The cover shall fit over the brush-tip ferrule by means of a friction fit or threaded connection.

3.7.7 *Physical requirements.*—Brushes shall conform to the applicable requirements of table I.

3.8 Workmanship.—Brushes shall be free from defects which affect their appearance or which might affect their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection lot.—For purposes of sampling and inspection, a lot shall consist of all brushes of one type and size manufactured under essentially the same conditions and presented for delivery at one time.

4.2 Sampling for lot acceptance.

4.2.1 *Sampling for visual and dimensional examination.*—Except when otherwise specified by Civil Government agencies, a random sample of brushes shall be selected from each inspection lot in accordance with table II with lot acceptance based on Military Standard MIL–STD–105, and subjected to the visual and dimensional examination specified in 4.3.

TABLE II.—*Sampling for visual and dimensional examination AQL (approx.) =2.5 percent defective*

Number of brushes in inspection lot	Number of brushes in sample	Acceptance number (defective)	Rejection number (defective)
65 and under	7	0	1
66 to 180	25	1	2
181 to 300	35	2	3
301 to 500	50	3	4
501 to 800	75	4	5
801 to 1300	110	6	7
1301 to 3200	150	8	9
3201 to 8000	225	11	12
8001 and over	300	14	15

4.2.2 *Sampling for lot acceptance tests.*—A random sample of brushes shall be selected from each inspection lot in accordance with table III and shall be subjected to the lot acceptance tests specified in 4.4.

TABLE III.—*Sampling for lot acceptance tests*

Number of brushes in inspection lot	Number of brushes in sample	Number of items nonconforming on any lot acceptance test.	
		Acceptance number	Rejection number
180 and under	1	0	1
181 to 500	2	0	1
501 to 800	3	0	1
801 to 1300	5	0	1
1301 to 3200	7	0	1
3201 to 8000	10	0	1
8001 and over	15	1	2

4.3 Visual and dimensional examination.—Each of the sample brushes selected in ac-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

cordance with 4.2.1 shall be visually and dimensionally examined to verify compliance with this specification as to material, workmanship, identification, finish of handle, depth and finish of ferrule, length of bristle, outside of ferrule, and other requirements not including tests. Any brush in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective brushes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Lot acceptance tests.—Each of the sample brushes selected in accordance with 4.2.2 shall be subjected to the applicable tests specified in 4.5 to verify compliance with this specification. Any sample brush which does not meet the requirements for any of these characteristics shall be rejected, and if the number of nonconforming brushes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Types "L" and "S" brushes shall be compared with the standard sample to determine conformance with 3.1.

4.5 Tests.

4.5.1 *Weight of bristles.*

4.5.1.1 *Removal of setting compound from bristles.*—The bristles and setting compound shall be removed from the brush. The setting shall be broken, without injury to the bristles, into small pieces approximately ½ by ¼ inch. The butt ends shall then be immersed in chloroform to a depth of not less than ⅛ inch above the setting compound. After soaking about 15 minutes, the bristles shall be removed from the chloroform, and the butt end grasped between the forefinger and thumb, and gently rolled back and forth. The setting compound will become loose and can be shaken off. This treatment shall be repeated until the setting compound has been entirely removed. After the setting compound has been removed, the bristles shall be conditioned as specified in 4.5.1.2.

4.5.1.2 *Conditioning of bristles.*—The bristles shall be conditioned for not less than 16 hours in air at a relative humidity of 50 ±4 percent, and at a temperature of 73° ±3° F. The weight of the bristles shall be determined in the same atmosphere to determine conformance with table I.

4.5.2 *Solvent resistance test (setting compound).*

4.5.2.1 *Composition of Solvent.*—The solvent shall be composed of the mixture shown in table IV.

TABLE IV.—*Composition of solvent*

	Percent by volume
Normal butyl acetate (Fed. Spec. TT–B–838)	15
Normal butyl alcohol (Fed. Spec. TT–B–846)	15
Methyl ethyl ketone (Fed. Spec. TT–M–261)	10
Toluol (Fed. Spec. TT–T–548)	60

4.5.2.2 *Test procedure.*—The brush shall be suspended so that the bristles and approximately one-half of the ferrule are immersed in chemically pure solvent for 1 hour at a temperature of 21° to 32° C. (70° to 90° F.). The brush shall be removed and examined.

4.5.2.2.1 *Observation.*—At the time of removal of the brush from the solvent, the setting compound shall be examined for excessive softening and loosening of the bristles. The brush shall then be air-dried for a period of 24 hours, the ferrule removed, and the setting compound examined for evidence of deterioration or loosening of the bristle. Any deterioration or loosening of bristles shall be cause for rejection.

4.6 Rejected lots.—Rejected lots may be resubmitted provided the contractor has removed or repaired all nonconforming brushes. Brushes shall again be selected and inspected to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment (Civil agencies only).

5.1 Packaging and packing.

5.1.1 *Nonmilitary packaging.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

5.1.1.1 *Levels A and B (unit packaging).*

5.1.1.1.1 *Types "L" and "S".*—Each brush shall be individually wrapped (bristle end) with suitable kraft paper. Wrapping shall be secured by rubber band or tape.

5.1.1.1.2 *Type "F".*—Each brush, complete with cover when specified, shall be individually packaged using paperboard folding, set-up, or metal-stayed boxes conforming to Federal Specifications PPP–B–566, PPP–B–676, PPP–B–665 or Military Specification MIL–B–4229, respectively. Boxes shall be closed and sealed in conformance with applicable container specification and appendices thereto. Each box shall contain at least one ball of naphthalene conforming to Federal Specification R–N–91, or other suitable moth repellant.

5.1.1.1.3 *Tips, replacement, type "F".*—Each tip shall have the bristles protected by insertion in a paperboard open-end tube of appropriate diameter and length.

5.1.1.2 *Intermediate packaging.*

5.1.1.2.1 *Types "L" and "S".*—Twelve brushes of one type and size, each wrapped as specified in 5.1.1.1, shall be packaged in paperboard folding, set-up or metal-stayed boxes conforming to Federal Specifications PPP–B–566, PPP–B–676, PPP–B–665 or Military Specification MIL–B–4229, respectively, with closure in accordance with the appendices thereto, as applicable. Brushes shall be arranged in boxes in reversing end to end order. Each box shall contain not less than three balls of naphthalene conforming to Federal Specification R–N–91 or other suitable moth repellant.

5.1.1.2.2 *Type "F".*—When specified, unit packaged brushes shall be packaged twelve each in a snug-fitting fiber box conforming to Federal Specification PPP–B–636 type I or II, class 1, closed and sealed in accordance with the appendix thereto. Additionally, all joints and seams including manufacturer's joints, shall be covered with tape in accordance with par. 30.2.2.2 (and figure 14) of PPP–B–636 (Appendix) except that, when used, gummed tape shall be not less than 3 inches in width.

5.1.1.2.3 *Tips, replacement, type "F".*—Twelve tips prepared as in 5.1.1.1.3 shall be packaged in boxes as specified in 5.1.1.2.1 for type "L" and "S" brushes, and shall contain naphthalene balls as specified therein.

5.1.1.3 *Level C.*—The brushes shall be packaged to afford protection against deterioration and damage in commercial quantities and in accordance with the manufacturer's commercial practice.

5.1.2 *Nonmilitary packing.*

5.1.2.1 *Levels A and B.*

5.1.2.1.1 *Types "L", "S", and replacement tips.*—Twelve intermediate boxes of 12 units each (144 brushes or 144 tips) packaged as in 5.1.1.1.1 or 5.1.1.2 as applicable, and of the same description, shall be packed in new shipping containers conforming to Federal Specification PPP–B–636, type I or II, class 1, style RSC. Board shall test not less than 200 p. s. i. and closure shall be in accordance with Federal Specification PPP–B–636 (Appendix).

5.1.2.1.2 *Type "F".*—Thirty-six units, packaged as in 5.1.1.1.2 (no intermediate packaging required) shall be packed in a new shipping container conforming to Federal Specification PPP–B–636 type I or II, class 1, style RSC. Board shall test not less than 200 p. s. i. and closure shall be in accordance with Federal Specification PPP–B–636 (Appendix).

5.1.2.2 *Level C.*—Brushes packaged for level A or C as specified (see 6.1) shall be packed in containers of the type, size, and kind commonly used for the purpose in a manner which will insure acceptance and safe delivery at destination. Shipping containers shall comply with the carrier rules and regulations as applicable to the mode of transportation.

5.1.3 *Military packaging.*

5.1.3.1 *Level A.*—Types "L" and "S" brushes shall be individually wrapped with kraft paper secured with rubber band or tape, and shall be packaged in quantities of 12 each arranged in reversing end to end order, and type "F" brushes individually packaged using paperboard folding cartons, setup, or metal-stayed boxes as the containers and conforming to Federal Specifications PPP–B–566, PPP–B–676, PPP–B–665 or Military Specification MIL–B–4229, respectively. The boxes shall be closed and sealed in conformance with the container specification and the appendices to the applicable container specification. Each box of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

brushes shall contain at least three balls of naphthalene conforming to Federal Specification R–N–91 or other suitable moth repellant. Replacement tips for type F shall have the bristle protected by insertion in a paperboard open-end tube of appropriate diameter and length, and each container shall contain at least one ball of naphthalene conforming to Federal Specification R–N–91, or other suitable moth repellant.

5.1.3.2 *Level C.*—The brushes shall be packaged to afford protection against deterioration and damage in commercial quantities and in accordance with the manufacturer's commercial practice.

5.1.4 *Military packing.*

5.1.4.1 *Level A.*—Brushes of the same type, packaged in accordance with level A or C as specified (see 6.1), shall be packed in overseas type wood-cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiber, wood-cleated paper overlaid, or wood-cleated plywood boxes conforming to Federal Specifications PPP–B–591, PPP–B–621, PPP–B–585, PPP–B–636 class 3, Military Specification MIL–B–10377 or Federal Specification PPP–B–601, respectively, at the option of the contractor. Shipping containers shall have case liners conforming to Military Specification MIL–L–10547 and shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to Federal Specification PPP–B–636 class 3 may be omitted provided all joints, seams and corners of the boxes are sealed with tape conforming to Federal Specification PPP–T–60 (type I or II, class 1). Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.1.4.2 *Level B.*—Brushes of the same type, packaged in accordance with level A or C as specified (see 6.1) shall be packed in domestic type wood-cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiber, wood-cleated plywood or wood-cleated paper overlaid boxes conforming to Federal Specifications PPP–B–591, PPP–B–621, PPP–B–585, PPP–B–636 class 2, PPP–B–601 or Military Specification MIL–B–10377, respectively, at the option of the contractor. Box closures shall be as specified in applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitation and shall conform to the special requirements of the box specification.

5.1.4.3 *Level C.*—Brushes, packaged for level A or C as specified (see 6.1), shall be packed in containers of the type, size and kind commonly used for the purpose in a manner which will insure acceptance and safe delivery at destination. Shipping containers shall comply with the carrier rules and regulations as applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Nonmilitary agencies.*—In addition to any special marking specified in the contract or order, each unit, intermediate package, and shipping container shall be marked in accordance with Federal Standard No. 123.

5.2.2 *Military.*—In addition to any special marking required by the contract or order, or herein, interior and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, symbol and date of this specification.
(*b*) Type and size required (see 1.2).
(*c*) Where samples of brushes may be seen (see 6.2).
(*d*) When brush tip cover is required (see 3.7.6).
(*e*) Selection of applicable levels of packaging and packing (see 5.1).
(*f*) Procurement of brush tips (see 6.4).

6.2 Standard sample.—The standard sample is reasonable facsimile of the master standard sample on file at the National Bureau of Standards, Washington, D. C. It is not intended that the master standard sample shall be used for the purpose of purchase or inspection of brushes, but that the master standard sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

will be used by Government agencies for comparison purposes to obtain a standard sample for their own use in purchase and inspection.

6.3 When specified by the procurement office, each bidder shall submit two brushes of each size that he proposes to furnish. The brushes delivered should be equal to the sample furnished with bids in respect to workmanship and balance, stiffness, solidity, and quality of bristle; and in any other respects not specifically covered in this specification.

6.4 When brush tips, 3.7.5, are procured separately, procurement officers shall requisition tips which are equal to those furnished by the manufacturer of the respective valve housing, 3.7.3.

6.5 Transportation description.—The transportation description applicable to this item is:

Brushes, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.
Motor volume minimum weight 24,000 pounds.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented inventions that may in any way be related thereto.

MILITARY INTEREST:
Army—Q E
Navy—Sh MC Or Y
Air Force

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.